



14-FMEA-FAILURE-CATALOG.md — 40 ways AURA breaks, and the guard for each



You cannot make software error-free by hoping. You make it error-free by writing down every way it can fail, assigning each one a detection and a guard, and then **injecting the failure in CI to prove the guard works**. This is that list.

1. Scoring

$RPN = \text{Severity} \times \text{Occurrence} \times \text{Detection}$ (each 1–5; Detection 5 = we would not notice). Anything $RPN \geq 45$ must have an automated guard before the owning phase can exit. Anything ≥ 80 blocks the release.

2. Import and IO

ID	Failure	Effect	S·O·D	Guard	Injected by
F-01	Card unplugged mid-import	Partial catalog, half-copied file	$5 \cdot 3 \cdot 2 = 30$	Copy to <code>.part</code> • fsync + atomic rename; checksum verify; resume from manifest	Fault injector unmounts volume
F-02	Truncated / bit-rotted RAW	Decoder crash or garbage pixels	$4 \cdot 3 \cdot 4 = 48$	Header + length validation before decode; quarantine <code>AURA-RAW-2004</code>	Corrupt corpus in <code>wedding-c-disaster</code>
F-03	Duplicate filenames from two bodies	Overwritten import	$5 \cdot 4 \cdot 3 = 60$	Identity = content hash + camera serial, never filename	Fixture with colliding <code>IMG_0001.CR3</code>
F-04	Network/NAS drive stalls	Run hangs forever	$4 \cdot 3 \cdot 4 = 48$	Per-IO deadline + watchdog;	Latency-injecting FUSE

ID	Failure	Effect	S·O·D	Guard	Injected by
				AURA-IO-1007 , retry then pause	mount
F-05	Disk fills during export	Half-written gallery	$5 \cdot 3 \cdot 2 = 30$	Pre-flight free- space estimate $\times 1.3$; atomic writes; AURA- EXPORT-8002	Small tmpfs target
F-06	Non-ASCII / emoji / 260- char paths	Windows failures at 90 % progress	$4 \cdot 3 \cdot 3 = 36$	\\?\ long- path prefix; NFC normalisation; path fixture suite	Unicode path fixtures
F-07	Two AURA instances, one catalog	Corruption	$5 \cdot 2 \cdot 4 = 40$	Advisory lock file with PID + heartbeat; second instance opens read-only	Dual-process test
F-08	Antivirus locks a file mid-write	Sporadic Windows failures	$3 \cdot 3 \cdot 4 = 36$	Sharing- violation retry ladder; documented exclusion guidance	File-lock injector

3. Catalog and state

ID	Failure	Effect	S·O·D	Guard
F-09	Power loss mid- transaction	Corrupt catalog	$5 \cdot 2 \cdot 2 = 20$	WAL + synchronous=FULL on checkpoint boundaries; integrity check on open
F-10	Migration fails halfway	Unopenable catalog	$5 \cdot 2 \cdot 3 = 30$	Auto-backup before migration; single transaction; rollback + AURA-DB-3005
F-11	Sidecar and DB disagree	Ghost edits	$4 \cdot 3 \cdot 4 = 48$	DB is authoritative; sidecar carries a generation counter; conflict → review queue

ID	Failure	Effect	S·O·D	Guard
F-12	Catalog on a synced folder (Dropbox/iCloud)	Corruption, phantom locks	$5 \cdot 3 \cdot 4 = 60$	Detect sync roots at project creation and refuse with guidance
F-13	SQLite BUSY storm under 32 threads	Stalls	$3 \cdot 4 \cdot 3 = 36$	Single writer actor; readers via WAL; bounded queue

4. Compute: GPU, ML, memory

ID	Failure	Effect	S·O·D	Guard
F-14	GPU device lost / driver TDR	Run dies	$5 \cdot 3 \cdot 2 = 30$	Recreate device once, then CPU fallback; keep tiles ≤ 2 s to avoid TDR
F-15	VRAM exhaustion on 61 MP files	Crash on the customer's exact camera	$5 \cdot 4 \cdot 3 = 60$	Tile-size solver from probed VRAM; halve-and-retry ladder; hard ceiling in 17-RESOURCE-BUDGETS.md
F-16	Host RAM OOM at 4,000 files	OS kills the app	$5 \cdot 3 \cdot 3 = 45$	Bounded channels, streaming decode, cache with a byte budget not an item count
F-17	Model file corrupt or tampered	Garbage output or code execution	$5 \cdot 2 \cdot 3 = 30$	Signature + SHA verify before load; refuse unsigned; AURA-ML-5001
F-18	EP silently falls back to CPU	8-minute cull becomes 90 minutes	$3 \cdot 4 \cdot 4 = 48$	Assert active EP after session creation; surface it in the UI; telemetry counter
F-19	Quantised model flips decisions near thresholds	Different gallery per machine	$4 \cdot 3 \cdot 4 = 48$	Parity gate in 12-DETERMINISM-CONTRACT.md ; hysteresis band around thresholds
F-20	Thermal throttling on laptops	Budget misses look like regressions	$2 \cdot 4 \cdot 3 = 24$	Record thermal state in benchmarks; compare only like-for-like
F-21	NaN/Inf propagates through the develop graph	Black or magenta pixels	$5 \cdot 2 \cdot 4 = 40$	Debug-build NaN guards per stage; release-build output sanity check before write

5. AI decision quality

ID	Failure	Effect	S·O·D	Guard
F-22	Culler deletes the only shot of the first kiss	Unrecoverable client harm	$5 \cdot 3 \cdot 3 = 45$	Story coverage guard: named moments can never reach zero; hard-block export
F-23	Face recognition merges two guests	Wrong person in album	$4 \cdot 3 \cdot 3 = 36$	Conservative merge threshold; cluster review UI; never auto-merge below 0.98
F-24	Retouch alters identity (slimming, skin lightening)	Ethical breach, brand damage	$5 \cdot 2 \cdot 2 = 20$	Hard-coded prohibition; geometry-change detector on faces; CI test asserts landmark drift ≈ 0 and skin-tone hue shift within $\pm 2^\circ$
F-25	Generative cleanup invents a limb / a guest	Fabricated reality	$5 \cdot 3 \cdot 2 = 30$	Masked inpainting only, area cap, no-face/no-hands rule, before/after diff bounded
F-26	Style profile overfits to 30 samples	Bizarre grades on new scenes	$4 \cdot 3 \cdot 4 = 48$	Minimum sample count per scene class; shrinkage to global prior; out-of-distribution detector
F-27	Confidence is miscalibrated (0.99 means 80 %)	Autonomy bands lie	$5 \cdot 3 \cdot 4 = 60$	Reliability diagrams per model; $ECE \leq 0.03$ gate; recalibrate on every release
F-28	Gallery consistency engine over-normalises	Every photo looks flat and identical	$4 \cdot 3 \cdot 3 = 36$	Bounded correction magnitude; variance floor; blind A/B panel
F-29	QC agent loops re-editing forever	Run never ends	$4 \cdot 3 \cdot 2 = 24$	Max 2 re-edit passes per photo; must show measurable improvement or escalate

6. Cloud, security, privacy

ID	Failure	Effect	S·O·D	Guard
F-30	API key leaks into a log or crash report	Financial + trust damage	$5 \cdot 2 \cdot 4 = 40$	Key only in OS keychain; redaction filter on every sink; CI secret-scan on fixtures
F-31	Client faces uploaded without	Legal exposure	$5 \cdot 2 \cdot 3 = 30$	Per-project consent flag gates the gateway; offline

ID	Failure	Effect	S·O·D	Guard
	consent			default; upload counter in run report
F-32	Runaway cloud spend	Surprise bill	$4 \cdot 3 \cdot 2 = 24$	Hard cap per project and per day; governor refuses past cap (AURA-CLOUD-6003)
F-33	LLM returns malformed or prompt-injected JSON	Bad decisions applied	$4 \cdot 3 \cdot 3 = 36$	Strict schema validation, value clamping, allow-list of actions; treat EXIF/filenames as untrusted data
F-34	Telemetry ships PII	Privacy breach	$5 \cdot 2 \cdot 4 = 40$	Schema allow-list only; no paths, no names, no thumbnails; opt-in

7. Release and lifecycle

ID	Failure	Effect	S·O·D	Guard
F-35	Auto-update mid-wedding run	Interrupted delivery	$4 \cdot 3 \cdot 2 = 24$	Updates deferred while any run is active; never restart without consent
F-36	New model version silently changes old galleries	Re-export differs from what the client saw	$5 \cdot 3 \cdot 3 = 45$	Projects pin their model set; upgrading is an explicit, logged action
F-37	Installer unsigned / notarisation fails	Customers blocked	$4 \cdot 2 \cdot 2 = 16$	Signing + notarisation verified in CI on a clean VM
F-38	Crash reporter itself crashes	No forensics	$3 \cdot 2 \cdot 5 = 30$	Out-of-process reporter; write minimal breadcrumb first
F-39	Licence check fails offline	Photographer locked out at a venue	$5 \cdot 3 \cdot 2 = 30$	30-day offline grace; never block an in-progress run
F-40	Uninstall deletes user galleries	Catastrophic data loss	$5 \cdot 1 \cdot 2 = 10$	App data and user data strictly separated; uninstall touches app data only

8. The chaos suite

`tests/chaos/` runs nightly against the 3,000-file synthetic wedding with the fault injector enabled:

1. **Kill -9 ladder** — kill the process at 5 %, 25 %, 50 %, 75 %, 95 % progress; each restart must resume and finish with an identical final gallery.
2. **Resource squeeze** — run with 4 GB RAM cap and a 2 GB VRAM cap; must complete via fallbacks, slower but correct.
3. **Hostile filesystem** — latency, EIO bursts, permission flips, disappearing volumes.
4. **Hostile cloud** — 429s, 500s, 30 s timeouts, malformed JSON, injected instructions in EXIF fields.
5. **Corrupt corpus** — 5 % of files truncated, wrong extension, zero-byte, or valid-header-garbage-body.
6. **Clock chaos** — NTP jumps backwards mid-run; DST boundary; cameras with wrong dates.

Exit condition for all six: **zero crashes, zero corruption, an accurate quarantine report, and a resumable catalog.**

9. Acceptance criteria

- ☐ Every RPN ≥ 45 row has a linked automated test that fails when the guard is removed.
- ☐ `tests/chaos/` green nightly on Windows and macOS.
- ☐ The kill -9 ladder produces identical final galleries at all five kill points.
- ☐ No entry in this table is marked "manual review" as its only detection.

10. Claude Code prompt

Act as `QAL` (QA Lead — Automation). Build `crates/aura-faults` with env-var-driven injection for the failure IDs in `14-FMEA-FAILURE-CATALOG.md`, then implement the six chaos scenarios in section 8. For each RPN ≥ 45 row, write the guard test and prove it by deleting the guard and showing the test fails. Report a table of failure ID → test name → PASS/FAIL with command output. Do not weaken any guard to make a test pass.